Junhee **Cho**

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Application for Ph.D Program

Dear Prof. Donguk Nam,

Who Am I

I am Junhee Cho, and I graduated with a Master's degree in Electrical Engineering from Pohang University of Science and Technology (POSTECH) in February 2026. I earned my Bachelor's degree in Electrical and Electronics Engineering from Hongik University, graduating with a GPA of 3.96/4.5 and a major GPA of 4.32/4.5. During my undergraduate studies, I researched GaN power devices and circuit simulations. At POSTECH, I was part of the Quantum Computing and Quantum Networks (QCQN) Laboratory, specializing in construction of cryogenic ion-trap quantum computing system, which requires highly reliable system design and implementation. As a researcher, I define myself as a persistent problem-solver, systematically overcoming recurring challenges in the laboratory and achieving the trapping of two ion qubits. My background in semiconductor design and quantum hardware has taught me to approach complex devices persistently and systematically, from design and integration to measurement and iteration.

What I Have Done

My research spans both semiconductor device optimization and quantum hardware development, providing me with a broad technical foundation. At Hongik University's Advanced Semiconductor Technology Lab (ASTL), I investigated the integration of depletion-mode AlGaN/GaN MOS-HFETs with clamp circuits to achieve normally-off operation. Using Silvaco-ATLAS, I developed a model that precisely matched real device behavior, converting it into a SPICE model for circuit simulation. This work resulted in a DC-DC boost converter with 99.5% power conversion efficiency, earning me the Undergraduate Poster Presentation Award at the 30th Korean Conference on Semiconductors.

Building upon this foundation, I shifted my focus toward quantum hardware at POSTECH. At the QCQN Laboratory, I spearheaded the design and fabrication of a cryogenic ion-trap system, successfully trapping two $^{40}Ca^{+}$ ion qubits for quantum computing research. Through this project, I gained hands-on experience with ion-trap fabrication and electrode plating. I also designed stable electrical signal delivery systems, built laser systems with low power fluctuation and tunable frequency, and constructed ultra-high vacuum (UHV) systems. This research was awarded the On-Site Outstanding Poster Award at the 31st Korean Conference on Semiconductors. Currently, I am investigating light-induced charging effects caused by oxide patches on electrodes. Based on surface photovoltage (SPV) theory, I am drafting a research paper on photon-induced surface charge dynamics, ultimately aiming to suppress surface-induced stray electric fields and enhance the long-term stability of trapped-ion qubits.

What I Want To Achieve

My previous research experiences have equipped me with a versatile skill set in semiconductor simulation and quantum hardware development. Drawing upon these experiences, I am certain that my acquired expertise will help me become productive within Q-SPIN lab, contributing positively to your research efforts.

As photonic integrated circuits (PICs) are being developed, the demand for compact and controllable on-chip light sources is increasing. However, realizing on-chip lasers that simultaneously satisfy process compatibility, performance, and long-term stability remains a key bottleneck across many platforms. In this context, I was particularly drawn to your group's work on GeSn/group-IV nanowire cavities and mid-infrared nanowire lasers, as well as active tuning enabled by nanomechanical oscillators and strain engineering, which I believe represents an important route toward silicon-

compatible integrated light sources. Through trapped ion-based light–matter interaction experiments, I have experienced that laser stability, noise, and drift control are decisive for reproducibility and long-duration operation. Building on this experience, I hope to contribute to making the tunability of nanowire lasers reliable under practical operating conditions. More broadly, I hope to reduce phase noise and long-term drift, and to establish actuation and control methods that can stably maintain a tuned state, ultimately enabling tunable on-chip lasers capable of sustained operation. In particular, I am interested in how strain-based actuation and device design can be engineered to achieve wide tunability while maintaining long-term frequency stability and spectral purity in GeSn nanowire lasers.

In the short term, I hope to quickly acquire the fabrication and characterization skills used in your lab and contribute meaningfully to ongoing research projects. At the same time, I would like to strengthen my physical understanding and modeling foundation so that I can interpret experimental results quantitatively. In the long term, I would like to help advance GeSn/group-IV nanowire lasers toward practical on-chip sources by lowering the lasing threshold and improving efficiency through loss reduction, mode engineering, and optimized coupling structures. It would truly be an honor to have the opportunity to join your research group and receive your guidance. Thank you very much for your valuable time and consideration, and I sincerely hope to be given this opportunity.

Sincerely,
Junhee Cho

Attached: Curriculum Vitae